ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 2 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 2 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 C ✓ 1.1.4 A ✓ 1.1.5 B ✓ 1.1.6 D ✓ 1.1.7 A ✓ 1.1.8 C ✓ 1.1.9 C ✓ 1.1.10 A ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 G ✓ 1.3.2 J ✓ 1.3.3 A ✓ 1.3.4 E ✓ 1.3.5 B ✓ 1.3.6 F ✓ 1.3.7 D ✓ 1.3.8 I ✓ 1.3.9 C ✓ 1.3.10 H ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Dull – boring, repetitive work, e.g. packing, machine tending ✓; dirty – unpleasant or unhealthy conditions, e.g. spray painting, foundry, sewer inspection ✓; dangerous – risk of injury, e.g. welding, handling hot metal, bomb disposal, heavy lifting ✓. Other reason (any ONE) ✓: better precision and consistency (quality); higher speed / productivity; works 24 hours; lower long-term cost (4)

2.2 SCARA: four axes, two parallel rotary joints plus vertical Z and wrist rotation ✓; cylinder / kidney-shaped envelope in a horizontal plane ✓; light to medium payload ✓; fast assembly, pick and place, screw driving ✓. Articulated: six rotary joints ✓; large, roughly spherical envelope that can reach around objects ✓; medium to very heavy payload ✓; welding, painting, palletising, machine tending ✓ (8)

2.3 (a) electromagnet / magnetic gripper ✓ – steel is magnetic, quick pick and release ✓ (b) soft / adaptive (bellows or multi-finger) gripper or vacuum ✓ – gentle, spreads the force so fruit is not damaged ✓ (c) 3-jaw (or adaptive) gripper ✓ – centres round parts of different diameters ✓ (6)

2.4 Point-to-point: only the end positions are controlled, the path between is not ✓ – spot welding / pick and place ✓; continuous path: the tool follows a controlled path all the way ✓ – arc welding / painting / glue dispensing ✓ (4)

[22]

QUESTION 3

3.1.1 $A_p = P - W_t$ ✓ $= 10 - 2{,}4 = 7{,}6$ kg ✓ (2)

3.1.2 No – 8 kg is more than the available payload of 7,6 kg ✓. (1)

3.2.1 $W = m \times g$ ✓ $= 6 \times 9{,}81 = 58{,}86$ N ✓ (2)

3.2.2 $T = W \times d$ ✓ $= 58{,}86 \times 0{,}4 = 23{,}54$ N·m ✓ (2)

3.3.1 $W = 2 \times 9{,}81$ ✓ $= 19{,}62$ N ✓ (2)

3.3.2 Friction must hold the weight: $F_{frict} = \mu \times F_{norm}$ ✓ $\rightarrow F_{norm} = 19{,}62 \div 0{,}4$ ✓ $= 49{,}05$ N; with safety factor 2: $49{,}05 \times 2$ ✓ $= 98{,}1$ N ✓ (accept per-jaw answers if clearly explained) (4)

[13]

QUESTION 4

4.1 Client-server: a central server provides services / data and clients request them ✓, easier central control and security ✓; peer-to-peer: all computers are equal and share resources directly ✓, no central server, cheaper but harder to manage ✓ (4)

4.2 Star: advantage – one faulty cable affects only one device / easy to add devices ✓; disadvantage – if the central switch fails the whole network stops / more cable ✓. Bus: advantage – little cable, cheap ✓; disadvantage – a break in the main cable stops the network / collisions, hard to find faults ✓ (4)

4.3 Gateway – connects networks that use different protocols (e.g. fieldbus to Ethernet / internet) ✓; router – forwards data between different networks and to the internet ✓; wireless access point – lets wireless devices join a wired network ✓ (3)

4.4 Any FOUR ✓✓✓✓: high speed; lower cost (standard hardware); easier maintenance and fault-finding; easy to extend; one network from sensors to office / cloud; rugged cables and connectors for harsh environments; real-time protocols available (4)

4.5 Advantages (any TWO) ✓✓: much less wiring; cheaper installation; diagnostics from each device; easy to add devices. Disadvantage (any ONE) ✓: more complex to set up; a cable fault can affect several devices; different fieldbus types are not compatible (3)

[18]

QUESTION 5

5.1 IoT devices collect data and send it over the internet to the cloud, where it is stored, processed and shown in apps; the cloud can also send commands back ✓✓. IIoT example: robot sensors send vibration / temperature data for predictive maintenance; production counts on a dashboard ✓ (3)

5.2 Any FOUR ✓✓✓✓: strong, changed default passwords; firewalls; separate the robot network from the office / internet (segmentation, VLAN); encryption; regular software / firmware updates; limit user access; VPN for remote access; staff training (4)

5.3.1 Robot: delta or SCARA robot ✓ – very fast pick and place from a conveyor ✓ (accept articulated with reason). Gripper: vacuum cups or multi-finger gripper picking several bottles at once ✓ – holds smooth bottles securely and quickly ✓ (4)

5.3.2 Any FOUR with purpose ✓✓✓✓: vision camera – locate bottles on the conveyor; photo-electric sensor – detect a box in position; safety fence / guard with interlocked door – stops the robot when opened; light curtain – stops the robot when a person enters; emergency stop – stop in an emergency; vacuum pressure switch – confirm a bottle is held; conveyor encoder – track bottle position (4)

5.3.3 Any TWO ✓✓: communication protocol between robot, PLC and conveyor; matching speeds / timing (conveyor tracking); data tracking and production records; space and work environment / safety layout; equipment and tools compatibility (2)

[17]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE